Opponent-Adjusted Evaluation of Pass Blocking and Pass Rushing Performance in the NFL

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Abstract

Evaluating offensive linemen and pass rushers at the player level is difficult because observable outcomes are sparse and heavily context dependent. Using 2021 regular-season Hudl tracking data, we construct a blocker–rusher interaction dataset and estimate two ridge-regularized Bradley–Terry paired-comparison models: a binary win/loss model aligned with the 2.5-second pass block win-rate definition and a multinomial severity model over `loss`/`win`/`hit`/`sack` with scalar outcome weights $(0, 0.10, 0.20, 1.00)$. The final dataset contains 153,138 interactions across 33,283 pass plays in 266 games. On a chronological 80/20 holdout split (n_{test} = 30,628), both models show consistent improvement over global and matchup baselines under log-loss evaluation, corresponding to relative log-loss reductions of about 0.2% to 1.2%. These validation gains remain directionally stable under game-level bootstrap resampling and are reinforced by external comparison to 2021 AP All-Pro selections, with the severity model showing the strongest alignment to expert recognition. Overall, regularized Bradley–Terry models provide an interpretable opponent-adjusted framework for evaluating NFL pass protection and pass rushing.

1 Introduction

Pass protection and pass rushing are central to modern NFL efficiency and roster construction, yet player-level evaluation in the trenches remains difficult. For rushers, box-score outcomes such as sacks and hits are rare and strongly mediated by context, including quarterback time-to-throw, coverage, and game situation. For blockers, the inverse problem holds: a lineman can execute consistently without generating a direct box-score statistic. Aggregate summaries therefore tend to blur individual ability with opponent quality, help structure, and team environment.

To better isolate line play, ESPN introduced pass block win rate (PBWR), which measures whether a blocker sustains a block for 2.5 seconds on a pass play (Burke, 2018). More recent tracking-based work has moved beyond sacks alone by modeling how pressure develops in space and time. For example, STRAIN summarizes how quickly defenders close space to the quarterback over time (Nguyen et al., 2023). These approaches provide richer information than sacks alone, but they are usually reported as season-level summaries rather than as jointly estimated, opponent-adjusted player ratings for blockers and rushers.

We study offensive and defensive line play within one opponent-adjusted paired-comparison framework. Each blocker–rusher engagement is treated as a matchup between a pass protector and a pass rusher, and we fit regularized Bradley–Terry (BT) models (Bradley and Terry, 1952; Friedman et al., 2010; Glickman and Jones, 2025). We estimate separate ridge-regularized BT models for the binary 2.5-second win/loss task and for an outcome-weighted severity task. Keeping the tasks separate allows task-specific validation, interpretation, and uncertainty quantification.

Contributions. Our main contributions are:

1. A paired-comparison framework that evaluates blockers and rushers jointly while preserving role-specific interpretation.
2. Separate regularized BT models for binary win/loss and multi-outcome severity, allowing each task to be evaluated on its own scale.
3. Chronological out-of-sample validation against task-specific baselines, including bootstrap uncertainty for predictive performance.
4. External validation against AP All-Pro selections using AUC and enrichment@K, compared with a raw win-rate baseline.
5. End-of-season leaderboards and cumulative weekly path uncertainty for longitudinal interpretation.

The remainder of the paper describes data and outcome construction, presents the modeling framework and validation design, and then reports internal validation, external validation, end-of-season leaderboards, and longitudinal uncertainty summaries.

2 Data and Outcome Construction

2.1 Tracking Data and Interaction Table

We use NFL player-tracking data from the 2021 regular season provided by Hudl. Tracking coordinates are recorded at 10 Hz and are accompanied by event annotations marking blocking engagements, pass attempts, and sacks.

We restrict attention to dropbacks by retaining plays containing either a forward pass event or a quarterback sack event. Within each retained play, we keep offensive linemen, identified pass rushers, and the quarterback. For each frame t, we compute the Euclidean distance between the quarterback and any non-quarterback player:

$$d_{p,t} = \sqrt{(x_{p,t} - x_{\mathrm{QB},t})^2 + (y_{p,t} - y_{\mathrm{QB},t})^2}. \tag{1}$$

Our unit of analysis is a blocker–rusher interaction, defined from the engagement labels in the tracking data. We generate a binary double-team indicator and set it to 1 when multiple blockers are assigned to the same rusher within overlapping time windows.

After filtering and labeling, the modeling table contains 153,138 blocker–rusher interactions across 33,283 pass plays in 266 games, with 620 rushers and 348 blockers. The double-team rate is 42.70%. This interaction table is the analysis sample for both BT models.

2.2 Outcome Definitions

All outcomes are coded from the rusher's perspective. We model two targets:

1. **Win/Loss target** (`win_target`): indicator of whether the rusher wins within the 2.5-second definition.
2. **Severity target** (`severity_outcome`): multinomial outcome in $\{\texttt{loss}, \texttt{win}, \texttt{hit}, \texttt{sack}\}$, with scalar severity mapping $0, 0.10, 0.20, 1.00$ for weighted evaluation.

The severity weights are anchored to EPA benchmarks for pass-rush outcomes reported in Eager (2018). Using the reported EPA benchmarks (no pressure $= 0.233$, hurry-only $= 0.019$, hit-only $= -0.161$, sack $= -1.856$), we rescale outcomes to the unit interval from the defender perspective using

$$w(o) = \frac{\text{EPA}_{\text{no pressure}} - \text{EPA}_o}{\text{EPA}_{\text{no pressure}} - \text{EPA}_{\text{sack}}}.$$

This mapping yields approximately $w(\texttt{win}) \approx 0.10$ and $w(\texttt{hit}) \approx 0.19$, and we use a rounded one-decimal scale for interpretability:

$$w(\texttt{loss}) = 0, \quad w(\texttt{win}) = 0.10, \quad w(\texttt{hit}) = 0.20, \quad w(\texttt{sack}) = 1.00,$$

so that severity scoring preserves ordering and is directly tied to observed EPA differentials.

We assign one outcome per interaction using severity priority `sack` > `hit` > `win` > `loss`. A `win` occurs if the rusher gains a decisive advantage within 2.5 seconds, operationalized as becoming closer to the quarterback than the blocker before the 2.5-second threshold. A `sack` is taken directly from event annotations. A `hit` is recorded when the quarterback is contacted behind the line of scrimmage before pass release without a sack. If none of these events occurs, the interaction is labeled `loss` from the rusher's perspective.

Observed sample frequencies in the full table are:

$$\hat{p}(\texttt{loss}) = 0.732, \quad \hat{p}(\texttt{win}) = 0.253, \quad \hat{p}(\texttt{hit}) = 0.0091, \quad \hat{p}(\texttt{sack}) = 0.0063.$$

3 Modeling Framework

We fit separate models because the binary and multiclass targets correspond to different estimands and are evaluated with different loss functions.

3.1 Win/Loss Ridge BT

For interaction t, with rusher $i(t)$, blocker $j(t)$, and double-team indicator D_t, the binary BT model is

$$\operatorname{logit} \mathbb{P}(Y_t = 1) = \alpha + r_{i(t)} - b_{j(t)} + \delta D_t,$$

where $Y_t = 1$ indicates rusher win.

We estimate parameters via ridge-penalized logistic regression:

$$\arg\min_{\theta} \left\{ -\ell(\theta) + \lambda \|\theta\|_2^2 \right\},$$

with λ selected by cross-validation on a log-spaced grid.

Throughout the paper, we report raw BT scores without affine rescaling so that figures and tables remain directly tied to fitted model coefficients.

3.2 Severity Ridge BT

For severity classes $c \in \{\texttt{loss}, \texttt{win}, \texttt{hit}, \texttt{sack}\}$, we fit a ridge-regularized multinomial BT model:

$$\mathbb{P}(C_t = c) = \frac{\exp(\eta_{t,c})}{\sum_{c'} \exp(\eta_{t,c'})}, \qquad \eta_{t,c} = \alpha_c + r_{i(t),c} - b_{j(t),c} + \delta_c D_t.$$

Intercept terms (α, α_c) absorb global outcome prevalence; player leaderboards are computed from non-intercept player effects.

The scalar severity prediction is the expected score under class probabilities, using the mapping `loss` $= 0$, `win` $= 0.10$, `hit` $= 0.20$, and `sack` $= 1.00$.

3.3 Chronological Split and Baselines

We sort interactions by `game_id`, `play_id`, and `event_game_index`, then apply a deterministic 80/20 chronological split (train = 122,510, test = 30,628).

Win model baselines:

1. **Global mean baseline**: train-set rusher win proportion.
2. **Matchup baseline (logit mean)**: smoothed rusher/blocker components combined on the logit scale with prior strength 25.

Severity model baselines:

1. **Global multiclass baseline**: train-set marginal class probabilities.
2. **Matchup multiclass baseline**: smoothed rusher and blocker class profiles (prior strength 50) combined through a multinomial logit mean.

3.4 Uncertainty

We report two complementary uncertainty procedures (Efron and Tibshirani, 1994):

1. **End-to-end bootstrap** ($B = 400$): resample games, refit with fixed λ selected from full-data CV, and recompute validation metrics and player ratings.
2. **Weekly path bootstrap** ($B = 100$): for each cumulative week checkpoint, resample past games and refit to obtain path-level uncertainty bands.

4 Results

We present internal predictive validation first, then external rank validation, followed by descriptive summaries of end-of-season ratings and weekly uncertainty.

4.1 Model Fit and Validation

Cross-validated ridge selected $\lambda_{\text{min}} = 1.31 \times 10^{-4}$ for the win model and $\lambda_{\text{min}} = 1.04 \times 10^{-4}$ for the severity model.

Table 1 reports holdout log loss relative to both global and matchup baselines. Both BT models improve on the corresponding baselines on the chronological test split. The bootstrap intervals from end-to-end game resampling are positive for both win-model comparisons and for severity versus global class frequencies; for severity versus the stronger matchup baseline, the interval overlaps zero, so that comparison should be interpreted as suggestive rather than conclusive.

Table 1: Out-of-sample log-loss validation. Improvement is baseline minus model, so larger values indicate better performance. The 95% intervals are percentile intervals from the game-level end-to-end bootstrap distribution of improvement and are therefore not constrained to be centered on the single-split estimate.

Task	**Baseline**	**Model log loss**	**Baseline log loss**	**Improvement**	**95% CI**
Win	Global	0.5568	0.5636	0.0068	[0.0045, 0.0093]
Win	Matchup	0.5568	0.5582	0.0014	[0.0004, 0.0024]
Severity	Global	0.6272	0.6347	0.0076	[0.0049, 0.0107]
Severity	Matchup	0.6272	0.6285	0.0014	[-0.0002, 0.0032]

4.2 External Validation Against All-Pro Selections

Having established holdout performance, we next ask whether the fitted rankings align with external expert evaluations. We compare BT rankings against 2021 AP All-Pro labels (first team and first+second team) and benchmark them against a simple empirical win-rate ranking baseline (rusher: $\mathbb{E}[\texttt{win\_target}]$, blocker: $\mathbb{E}[1 - \texttt{win\_target}]$). We report:

1. **Rank AUC**: probability that a randomly chosen All-Pro player is ranked above a randomly chosen non-All-Pro player. With labels $y_i \in \{0, 1\}$, scores s_i, $n_+ = \sum_i \mathbf{1}\{y_i = 1\}$, and $n_- = \sum_i \mathbf{1}\{y_i = 0\}$, we use the Mann–Whitney form

$$\mathrm{AUC} = \frac{1}{n_+ n_-} \sum_{i:y_i=1} \sum_{j:y_j=0} \Big[\mathbf{1}\{s_i > s_j\} + \tfrac{1}{2}\mathbf{1}\{s_i = s_j\}\Big].$$

2. **Enrichment@K**: precision@K divided by base rate, where K is the number of All-Pro positives in that role/accolade slice (equivalently, the AP selection-slot count for that slice):

$$\text{Enrichment@}K = \frac{\text{precision@}K}{n_+/n}, \qquad \text{precision@}K = \frac{\text{hits@}K}{K}.$$

Here n is the number of players in the role/accolade slice.

Table 2 shows that the severity model leads on AUC in three of four role/accolade slices and achieves the strongest enrichment@K for blockers (first+second team) and rushers (both accolade sets). The raw win-rate baseline attains moderate AUC in some slices but yields enrichment@$K = 0$ throughout, indicating that it fails to place any All-Pro selections inside the relevant top-K sets.

Table 2: All-Pro alignment metrics by model, role, and accolade set (2021 AP labels). K equals the number of positives in each role/set (the corresponding AP slot count). AUC is rounded to three decimals and Enrichment@K to two decimals.

Model	**Role**	**Accolade set**	K	**AUC**	**Enrichment@K**
Raw win-rate baseline	Blocker	First+Second Team	10	0.653	0.00
Raw win-rate baseline	Blocker	First Team	5	0.712	0.00
Raw win-rate baseline	Rusher	First+Second Team	14	0.718	0.00
Raw win-rate baseline	Rusher	First Team	7	0.702	0.00
Severity	Blocker	First+Second Team	10	0.876	13.92
Severity	Blocker	First Team	5	0.850	0.00
Severity	Rusher	First+Second Team	14	0.727	12.65
Severity	Rusher	First Team	7	0.771	25.31
Win/Loss	Blocker	First+Second Team	10	0.654	3.48
Win/Loss	Blocker	First Team	5	0.735	0.00
Win/Loss	Rusher	First+Second Team	14	0.768	6.33
Win/Loss	Rusher	First Team	7	0.749	0.00

4.3 Ratings and Leaderboards

After internal and external validation, we summarize the fitted end-of-season ratings. Figure 1 shows raw BT score distributions by role for both tasks. Figure 2 reports top players by model and role (minimum 200 interactions to reduce small-sample volatility). Scores are interpreted within each model (win versus severity) rather than across models, because the two targets induce different scales. The win/loss and severity leaderboards are directionally similar for elite rushers but differ more noticeably for blockers, which is consistent with the severity model's heavier emphasis on high-impact outcomes.

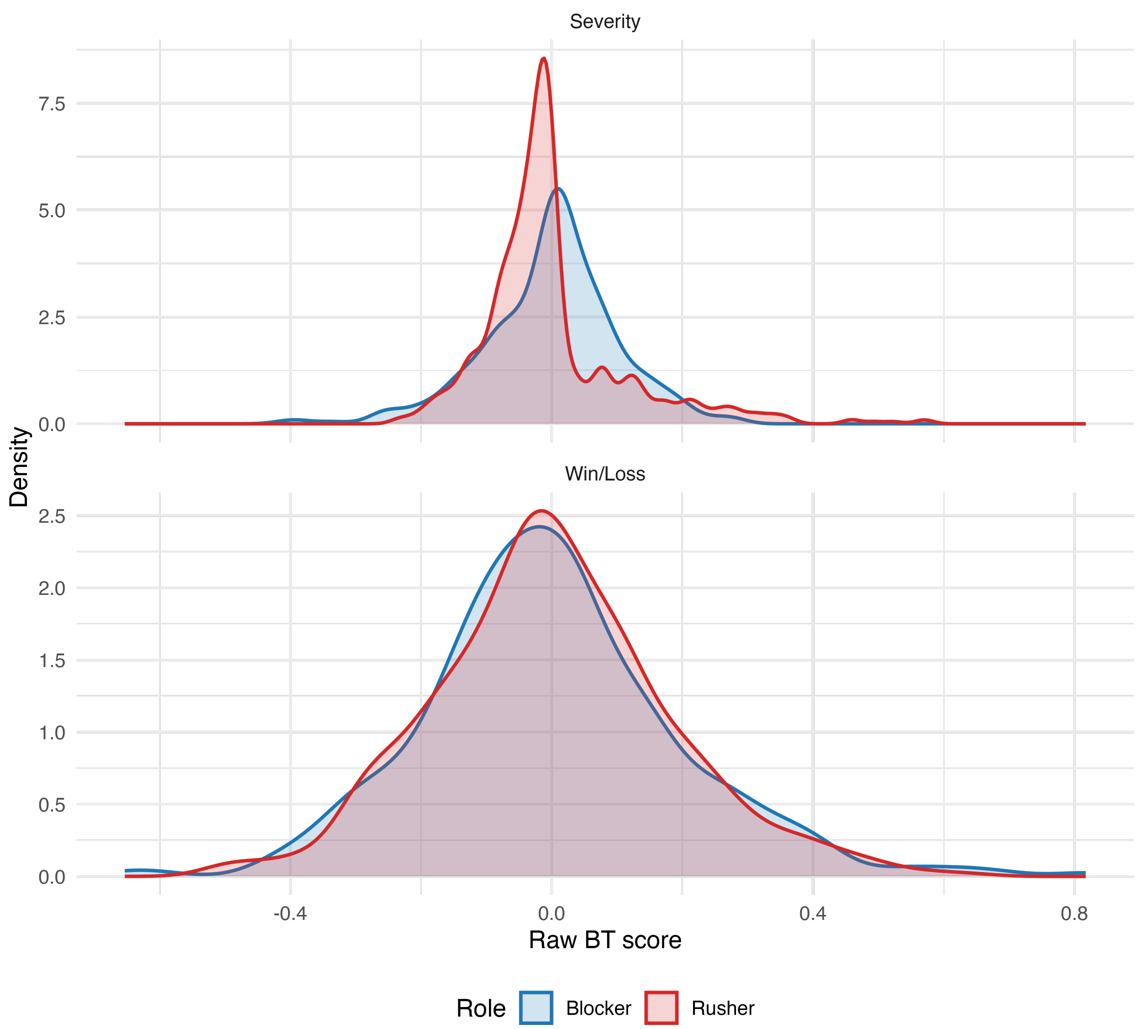


Figure 1: Distribution of raw BT scores by model and role.

4.4 Weekly Path Uncertainty

The weekly path bootstrap adds a longitudinal view to the end-of-season summaries. The regular-season dataset yields 18 cumulative weekly checkpoints for each player-model-role series. Figure 3 shows weekly trajectories for the top three players in each model-role panel, using the same minimum interaction threshold ($n \geq 200$) applied in the leaderboard visualizations.

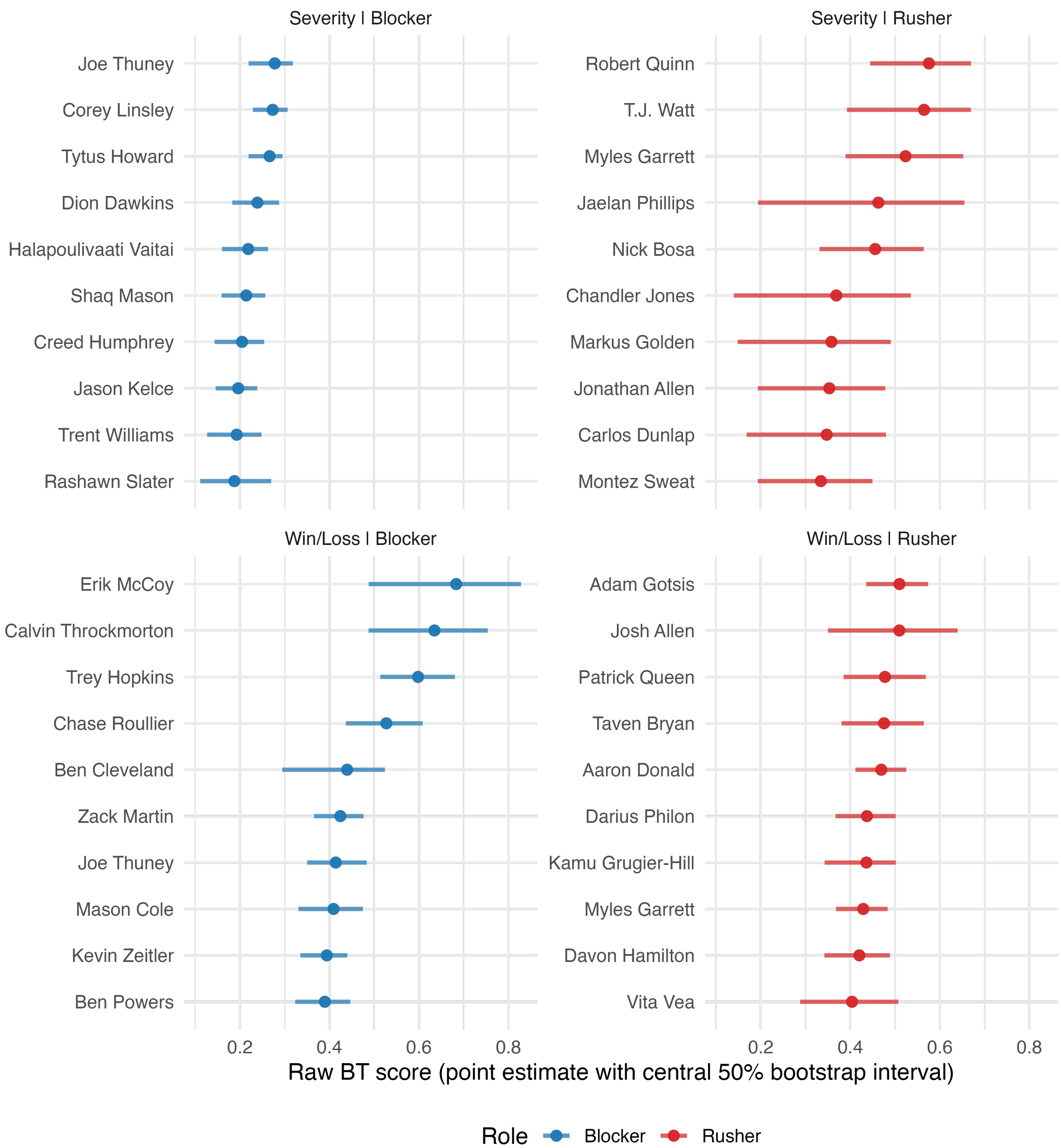
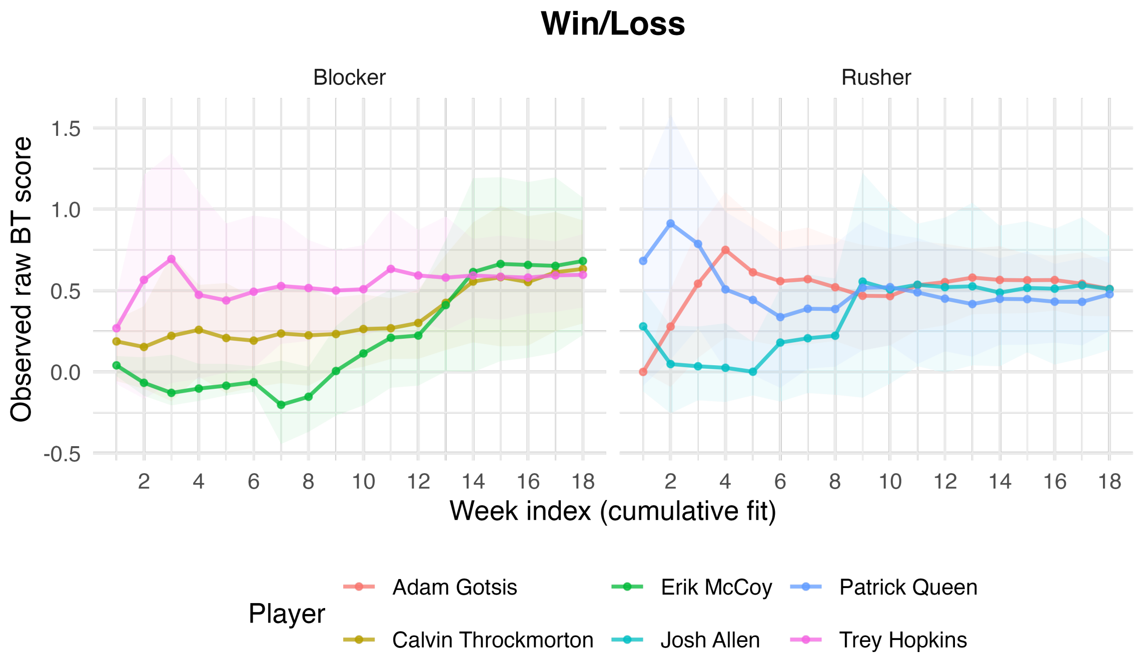


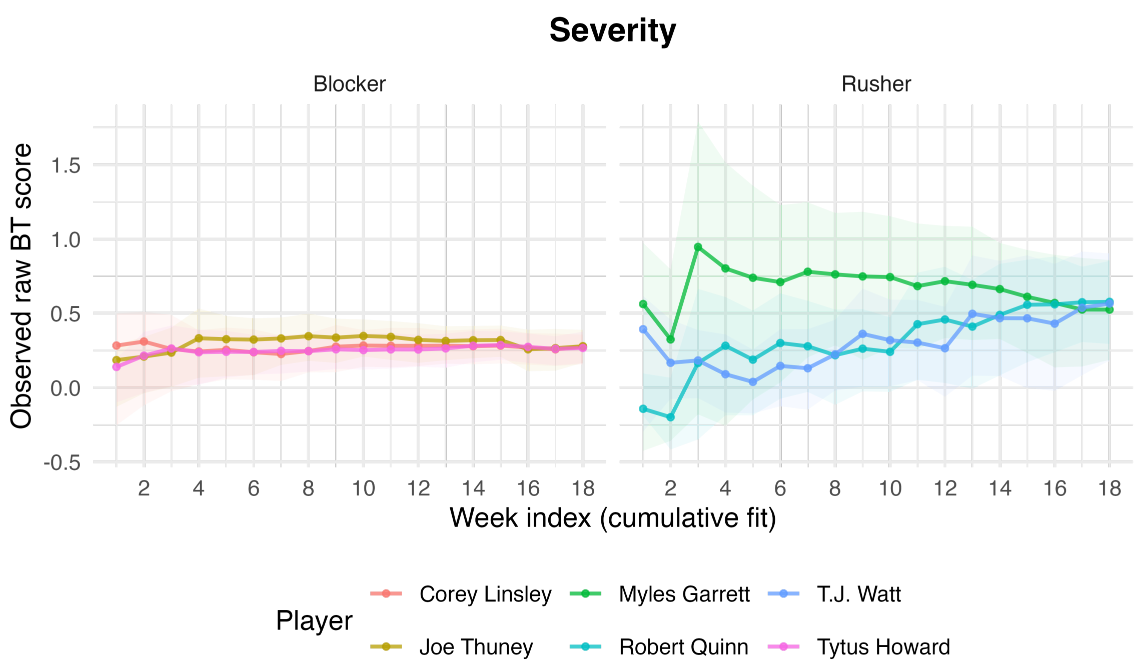
Figure 2: Top 10 players by raw BT score in each model-role panel (minimum 200 interactions). Points are estimates and horizontal bands are central 50% bootstrap intervals (25th–75th percentiles).

Table 3: Top five players by model and role (minimum 200 interactions). Values are raw BT scores.

Model	**Role**	**Player**	**Rating**
Win/Loss	Rusher	Adam Gotsis	0.510
Win/Loss	Rusher	Josh Allen	0.509
Win/Loss	Rusher	Patrick Queen	0.477
Win/Loss	Rusher	Taven Bryan	0.475
Win/Loss	Rusher	Aaron Donald	0.469
Win/Loss	Blocker	Erik McCoy	0.683
Win/Loss	Blocker	Calvin Throckmorton	0.635
Win/Loss	Blocker	Trey Hopkins	0.598
Win/Loss	Blocker	Chase Roullier	0.527
Win/Loss	Blocker	Ben Cleveland	0.440
Severity	Rusher	Robert Quinn	0.576
Severity	Rusher	T.J. Watt	0.565
Severity	Rusher	Myles Garrett	0.523
Severity	Rusher	Jaelan Phillips	0.462
Severity	Rusher	Nick Bosa	0.455
Severity	Blocker	Joe Thuney	0.278
Severity	Blocker	Corey Linsley	0.273
Severity	Blocker	Tytus Howard	0.266
Severity	Blocker	Dion Dawkins	0.239
Severity	Blocker	Halapoulivaati Vaitai	0.219



(a) Win/Loss model.



(b) Severity model.

Figure 3: Weekly cumulative raw BT score paths with bootstrap uncertainty ribbons (mean $\pm$ 1.96 SD) for the top three players in each model-role panel (minimum 200 interactions).

5 Discussion

5.1 Interpretation

The results support regularized BT as a practical opponent-adjusted framework for trench evaluation. Internal validation shows consistent improvement over global baselines and modest but mostly positive improvement over stronger matchup baselines. External validation is especially encouraging for the severity model, whose rankings concentrate 2021 All-Pro selections near the top of the list. Taken together, these results suggest that modeling richer pass-rush outcomes can sharpen separation among elite players while preserving an interpretable blocker–rusher comparison structure.

5.2 Limitations and Future Work

Several limitations suggest natural extensions. First, the `win` label is based on a distance-to-quarterback rule and may not fully capture functional pressure quality. More realistic labels could incorporate pocket geometry, rush lane, and quarterback decision constraints. Second, help mechanisms (chip blocks, tight-end/running-back assistance, and protection-slide structure) are only partially captured by a coarse double-team indicator. Third, sacks and pressure proxies are influenced by quarterback time-to-throw and play design, which are only indirectly represented. Finally, although BT adjusts for opponent strength, it does not explicitly model teammate effects or position-family hierarchy; hierarchical shrinkage and multi-season pooling are natural next steps.

6 Conclusion

Regularized BT models provide a stable and interpretable framework for NFL pass-rush and pass-protection evaluation. In this 2021 regular-season study, the models outperform simple baselines on both binary win/loss and outcome-severity tasks, while the severity formulation shows especially strong external alignment with All-Pro selections. These findings support opponent-adjusted paired-comparison methods as a useful basis for future tracking-based work on offensive and defensive line evaluation.

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